

FALLING BATTERY PRICES

○ As lithium prices declined to a low of \$10,150/tonne in September, lithium-ion battery prices for EVs have dropped

Vehicle	Battery Pack Size	2020 price	2024 price
Tata Tiago	19.2 kWh	₹1,77,400	₹1,25,800
Tata Nexon	30 kWh	₹2,77,200	₹1,96,560
Tata Nexon	40.5 kWh	₹3,74,200	₹2,65,350
Kia EV9	99.8 kWh	₹9,22,150	₹6,53,890
Ola S1	3 kWh	₹27,720	₹19,650

NOTE PRICES ROUNDED OFF; FOREIGN CURRENCY EXCHANGE RATE AT ₹84 PER \$)

SOURCE BENCHMARK MINERALS, BT RESEARCH

CHARGING AHEAD

○ Manufacturers assert that the cost-to-service of electric vehicles is significantly lower as they have fewer moving components

Vehicle Tech	Cost/km	Total running cost/month
EV (30 kWh, 300 km range)	₹1.38	₹1,380
Petrol (12 kmpl mileage)	₹7.89	₹7,890
Diesel (16 kmpl mileage)	₹5.84	₹5,840

NOTE ASSUMING AN AVERAGE TRAVEL OF 1,000 KM PER MONTH **SOURCE** NITI AAYOG

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per kWh was \$110. These prices are expected to decline even further, resulting in more affordable battery packs in the coming days.

According to Benchmark Minerals' research, prices of lithium-ion batteries have fallen 57% since 2016. During the same period, the market size of lithium-ion batteries has surged, multiplying by more than 20 times.

LITHIUM PRICES IN A TAILSPIN

Lithium-ion battery prices have dropped due to a significant decline in the prices of lithium, the wonder mineral powering the EV revolution. Lithium carbonate, the major source of the mineral, was at an all-time high of \$82,000 per tonne in December 2022. Since then, several factors have contributed to the decline in prices; it reached an all-time low of \$10,150 per tonne in September 2024.

According to a report by HSBC Global Research, lithium has little scope to rebound further, as "sup-

ply remains abundant relative to demand". In early September, there was a rally in lithium futures on news of supply cuts from Chinese battery maker CATL's mine closure and maintenance of production lines by lithium player Jiuling. However, the rally was short-lived.

ADVANTAGE EVs?

The falling cost of battery replacement is the cherry on top of the huge running cost advantage that EVs have over traditional internal combustion engine (ICE) technology. However, car manufacturers are not passing on the benefits of lower battery prices to buyers. This reflects in the fact that there has been no corresponding cut in sticker prices of EVs in the past few years.

A journey cost calculator on the NITI Aayog website highlights the massive advantage EVs have over petrol and diesel cars (see chart 'Charging Ahead'). Calculations have been made assuming an average travel of 1,000 km per month.

While owner accounts differ re-

garding service costs, manufacturers claim that the cost-to-service of EVs is much lower as they have fewer moving parts.

CHINA'S ROLE

The increasing adoption of EVs, especially in China, has pushed up the demand for lithium; but supply has been growing even faster. That, together with slowing EV sales outside of China, has pushed lithium prices down. A recent report from Goldman Sachs estimates that supply may exceed demand by 57% next year.

Countries like the US, Japan, India, China, the Netherlands, and many others have set ambitious targets for EV adoption. India, for instance, aims for 30% of new vehicle sales to be EVs by 2030. Although charging infrastructure and range anxiety continue to pose challenges in achieving this goal, the falling cost of battery packs is likely to entice more people to plug in. **BT**

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